Reusable Synthetic Test Data - BL-28

All names, roll numbers and emails are fictional. Use this data only in local development or an isolated test database; never substitute real student information.

Loading it

```
python manage.py migrate
python manage.py seed_test_data
```

The command creates everything below in one step. It runs only when `DEBUG=True` and refuses to run if the accounts already exist, so it cannot overwrite real data. To reload, use a fresh database. Its figures are checked by `apps/academics/test_seed_data.py`, so this document and the data cannot drift apart.

Every account's password is `Attend#2026` (no forced password change).

Two-factor login (BL-33) is on by default, so each account sets up an authenticator app at its first login. To explore the data quickly on your own machine, add `OTP_REQUIRED=False` to your local `.env` and restart the server.

Accounts

Role	Username	Scope
Admin	`ADMIN1`	All records
Faculty	`F001`	Teaches CS101 in Section A (students S001, S002, S004)
Faculty	`F002`	Teaches MA101 in Section B (students S001, S003)
Faculty	`F003`	No classes
Student	`S001`	CS101 (Section A) and MA101 (Section B)
Student	`S002`	CS101 (Section A)
Student	`S003`	MA101 (Section B); joined after its sessions, so **no attendance records**
Student	`S004`	CS101 (Section A)

Academic data

Record	Value
Sections	`B.Tech CSE Year 2` A and B
Subjects	`CS101` Programming, `MA101` Mathematics
Teaching assignments	F001 -> CS101 in A; F002 -> MA101 in B

Attendance

- **CS101, Section A:** 100 sessions from 1 to 17 June 2026 (periods 1-6 each day), recorded by F001.
- **MA101, Section B:** 3 sessions for S001 on 1 Oct (Present), 2 Oct (Absent) and 3 Oct 2026 (Present), period 1, recorded by F002.

Student	CS101	MA101	Overall	Below 75%?
S001	74/100 (74.00%)	2/3 (66.67%)	76/103 (73.79%)	**Yes** (both subjects and overall)
S002	75/100 (75.00%)	-	75/100 (75.00%)	No - equal is not below
S004	76/100 (76.00%)	-	76/100 (76.00%)	No
S003	-	N/A - no classes yet	N/A - no classes yet	No - zero classes are never flagged

Dashboard figures (threshold 75%)

Dashboard	Figures
Admin	4 students, 3 faculty, 2 sections, 103 sessions; 1 student below overall; 2 subject shortages (S001 CS101, S001 MA101)
F001	1 class, 3 students, 100 sessions; 1 below; 1 shortage (S001 CS101)
F002	1 class, 2 students, 3 sessions; 1 below; 1 shortage (S001 MA101)
F003	All zero
At threshold 80% (Admin)	3 students below overall (S001, S002, S004); 4 subject shortages

Date-filter examples (S001, inclusive dates)

Filter	Result
MA101, 2 Oct to 2 Oct 2026	0/1 (0.00%)
MA101, 2 Oct to 3 Oct 2026	1/2 (50.00%) - both boundary dates included
Any subject, 10 Oct to 11 Oct 2026	No records - "No attendance records match these filters."
Start date after end date	Rejected - "End date must be on or after the start date."

Traceability

- **BL-15 / FR-09 / UC-08:** own-student access, subject and date filters, inclusive boundaries, no-match state.
- **BL-22 / FR-13 / UC-10:** default 75, whole numbers 1-100, below/equal/above boundaries (S001/S002/S004), zero-record student (S003), Admin-only changes.
- **BL-25 / FR-17 / UC-12:** role-scoped dashboard figures, including a faculty member with no classes (F003).
- **BL-28:** the test plan's concrete cases use these accounts and figures.

Non-functional tests

Performance and capacity tests (test plan PT*01-PT*04) need a larger synthetic dataset - 1,000 students, 60 faculty and one semester of sessions - in a staging environment. This small dataset is for functional testing only.